



InkRush

ECE: 3330 Intro to Software Design

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PROJECT OVERVIEW

InkRush is a multiplayer Pictionary-style drawing and guessing game developed in Java using JavaFX, Socket networking(TCP), multithreaded server design, and a database-backed word system. Players connect to a central server, draw in real time, guess words, earn points, and rotate roles across multiple rounds.





NOTABLE FEATURES

Custom Avatars: Draw your own avatar in lobby

Confetti Physics: Animated victory screen

Smart Word System: No word repetition logic

Live Drawing Sync: Real-time stroke updates

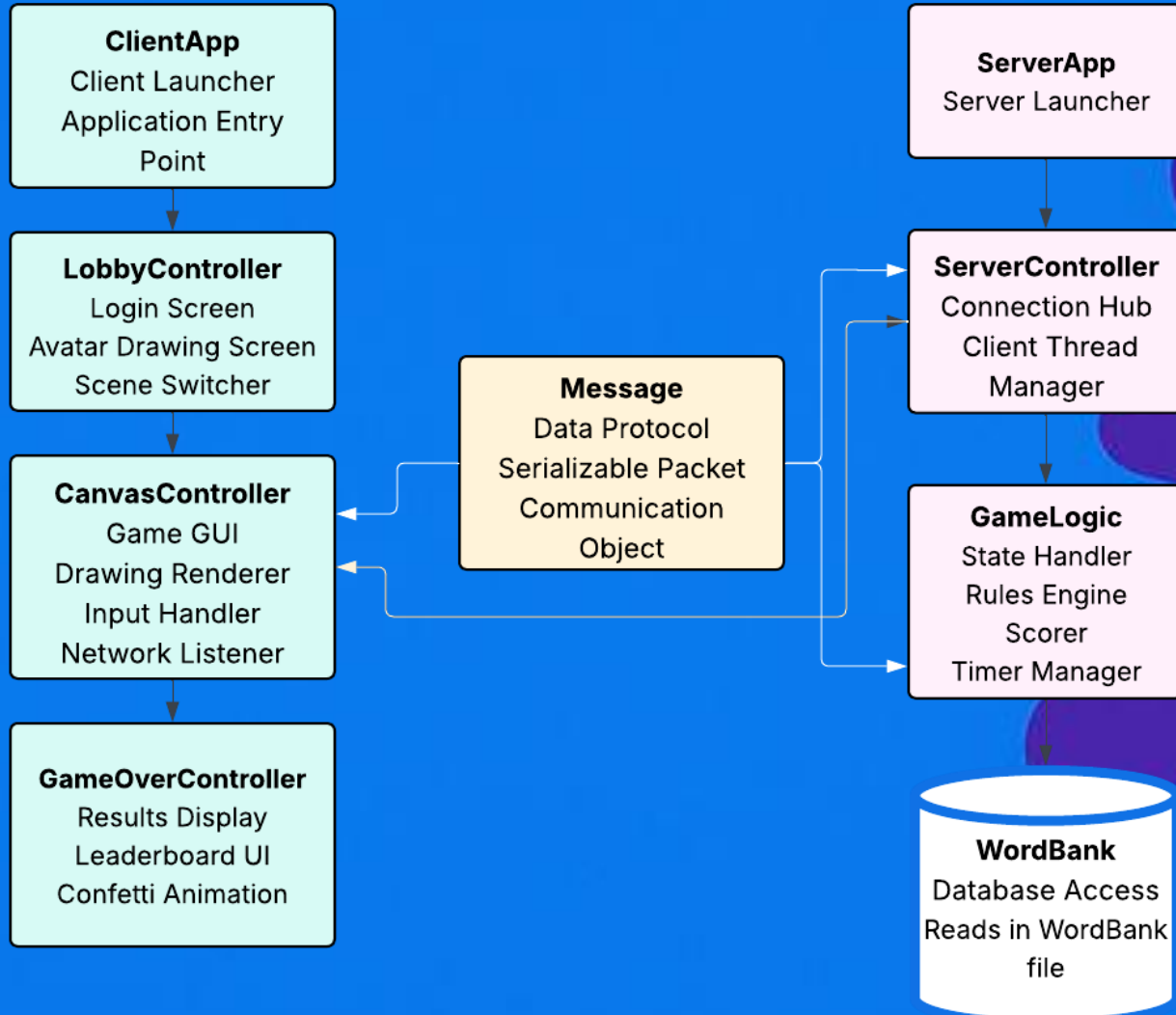
Audio Polish: Music and sound effects

Dynamic Scoring: Speed-based points system





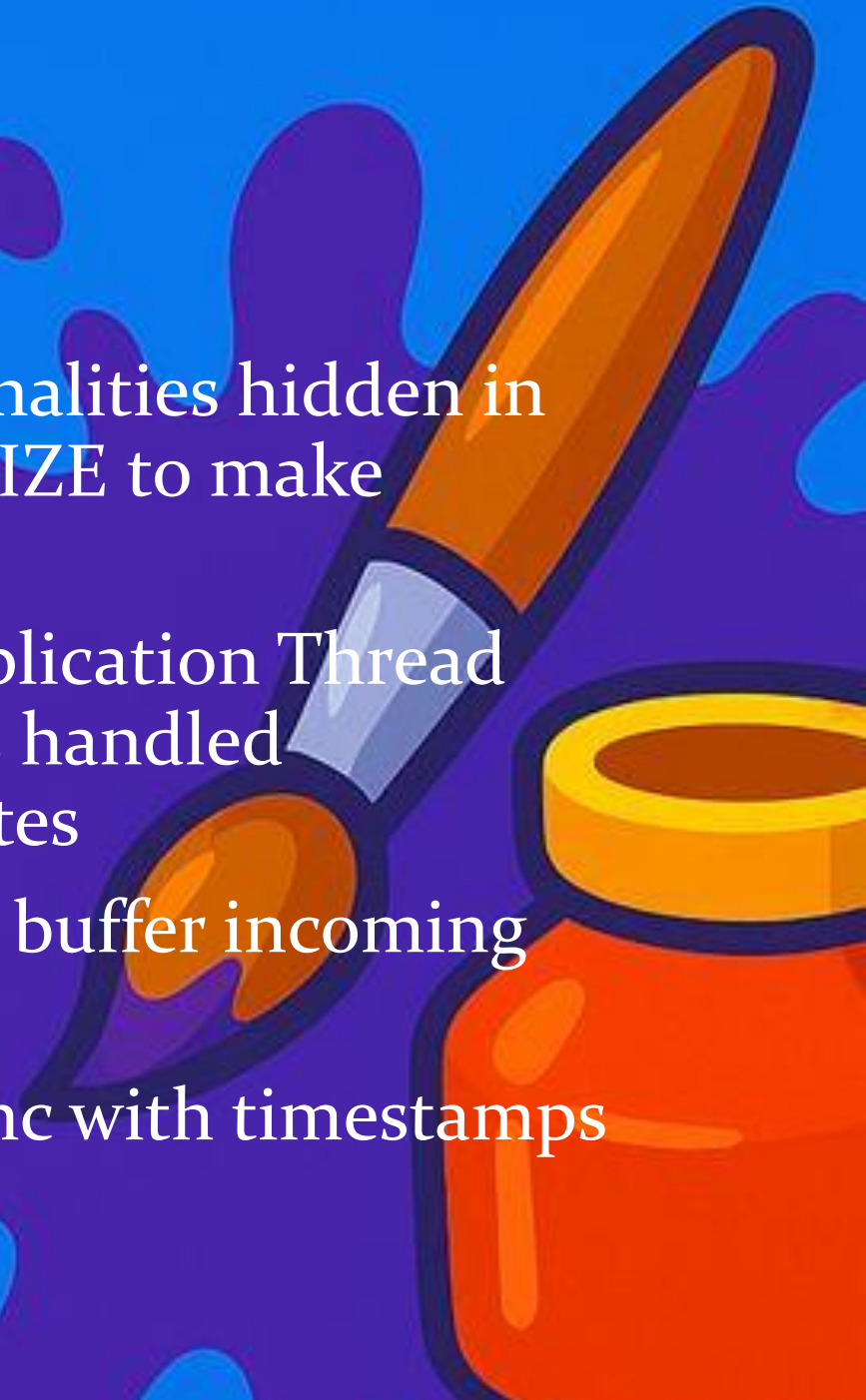
GENERAL STRUCTURE & GAMEFLOW





TECHNICAL CHALLENGES

- **Canvas Resizing:** Explored different functionalities hidden in SceneBuilder (ex. Using `USE_COMPUTED_SIZE` to make components stretch according to page size)
- **Thread Safety:** Ensured only the JavaFX Application Thread made GUI updates while background threads handled server/client communication and game updates
- **Drawing Synchronization:** Used a queue to buffer incoming packets, preventing data loss and lag.
- **Race Conditions in Scoring:** Server-side sync with timestamps



TEAM TAKEAWAYS

-  - **Nomi:** CLI (command line instructions) comes in clutch when trying to config the project which is developed locally on Linux machine
-  - **Jet:** Creating a serializable message class to act as the client/server protocol made handling game events easy
-  - **El:** Reading the textbook about multithreading in JavaFX was very helpful to understand how to implement it correctly
-  - **Amit:** Using SceneBuilder made designing the GUI much easier than Swing, and it was great learning about how Platform updates it safely

GAME DEMONSTRATION!!!

